

```

root_path = fileparts(mfilename("fullpath"));
root_path = pwd;
data_path = fullfile(root_path, 'sym4D_cutsAndFits');

if ~exist('cuts2fit400', 'var')
    ld = load(fullfile(data_path, 'multicuts_fit_dataGH_ei400meV.mat'));
    cuts2fit400 = ld.cuts2fit400;
end
if exist('used_ds400', 'var')
    used_ds = used_ds400;
else
    used_ds = containers.Map()
    sel_info = struct('cuts', [], 'en_range', [], 'fname', '', 'bg_par', []);
    %
    sel_info.cuts = cuts2fit400.cuts_dir001FFfit(1);
    sel_info.en_range = 50:10:160;
    sel_info.cut_dim = 2;
    sel_info.fname = 'EnFit_Ei400ref001off110_2D_constFFCut';
    sel_info.bg_par = {};
    used_ds('Ei400Br110dir001') = sel_info;
    %
    sel_info.cuts = cuts2fit400.cuts_dir001FFfit(2);
    sel_info.en_range = 50:10:200;
    sel_info.cut_dim = 2;
    sel_info.fname = 'EnFit_Ei400ref001off200_2D_constFFCut';
    sel_info.bg_par = {};
    used_ds('Ei400Br200dir001') = sel_info;

    %
    sel_info.cuts = cuts2fit400.cuts_dir001FFfit(1);
    sel_info.en_range = 50:10:160;
    sel_info.cut_dim = 1;
    sel_info.fname = 'EnFit_Ei400ref001off110_1D_constFFCut';
    sel_info.bg_par = {};
    used_ds('Ei400Br110dir001_1D') = sel_info;
    %
    sel_info.cuts = cuts2fit400.cuts_dir001FFfit(2);
    sel_info.en_range = 50:10:200;
    sel_info.cut_dim = 1;
    sel_info.fname = 'EnFit_Ei400ref001off200_1D_constFFCut';
    sel_info.bg_par = {};
    used_ds('Ei400Br200dir001_1D') = sel_info;
    used_ds400 = used_ds;
end

```

used_ds =

Map with properties:

```

    Count: 0
    KeyType: char
    ValueType: any

```

Show all accessible properties of Map

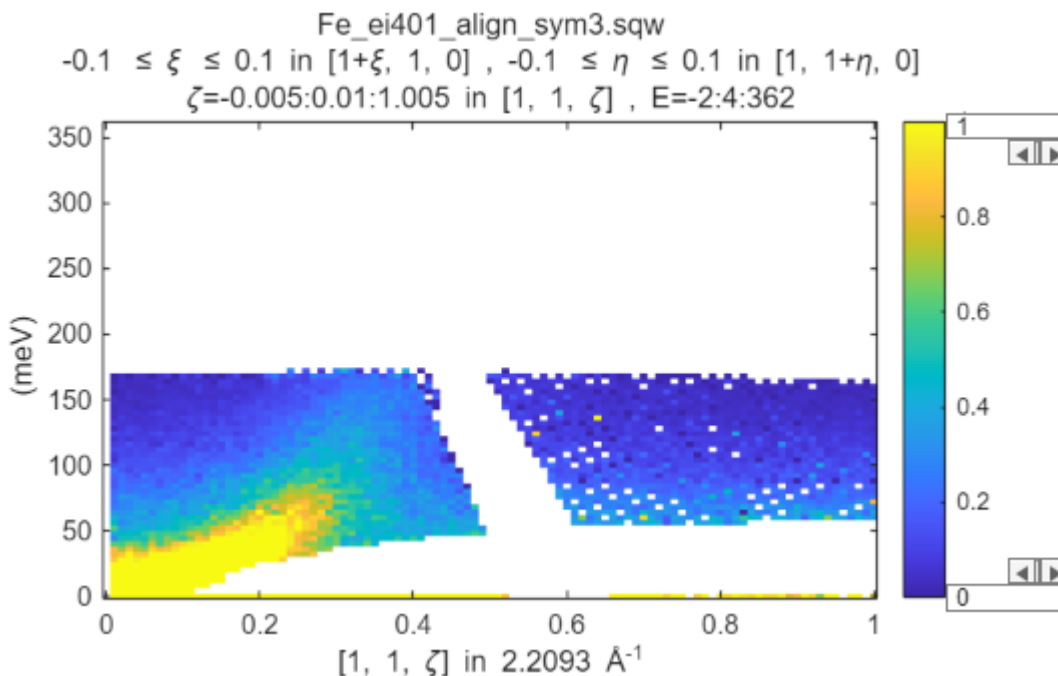
```

%cuts_list = cuts2fit400.cutsS_list; % all available cuts
%cuts_list = {cuts2fit400.cut_b200_200sel}; % cuts in low ff direction around
<200>
%cuts_list = cuts2fit400.cuts_dir001FFfit(1);
select= 'Fe400Br110dir001_1D';
sel_info = used_ds(select);

cuts_list = sel_info.cuts;

mi = maps_instrument(400,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for i=1:numel(cuts_list)
    cuts_list{i} = cuts_list{i}.set_instrument(mi);
    cuts_list{i} = cuts_list{i}.set_sample(sample);
    plot(cuts_list{i}); lz 0 1; keep_figure;
end

```



```

dir_name = "GH";

dE_step = 4; %original energy transfer step data were binned to. No point in
going finer
half_dE = 5; % half width of data binning
%cut_en = 50:10:260;
%cut_en = 50:10:160; %cuts2fit400.cuts_dir001FFfit(1);
cut_en = sel_info.en_range; %cuts2fit400.cuts_dir001FFfit(2);
cut_dimensions = sel_info.cut_dim;

%cl= {cuts_list{3}};
%cl = {w2_test_cut};
% if isfield(cuts2fit400,'bg_par_010off000')

```

```

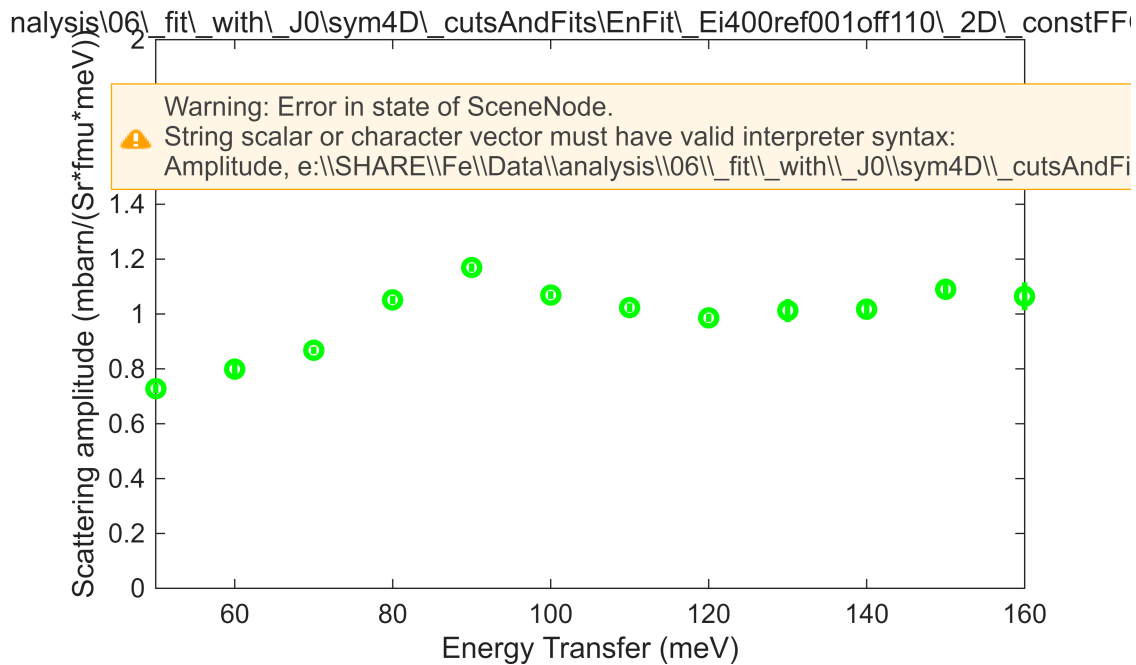
%     if numel(cuts_list) == 3
%         bg_par =
{cuts2fit400.bg_par_010off000.p,cuts2fit400.bg_par_010off100.p,cuts2fit400.bg_par
_010off200.p};
%     else
%         bg_par = {cuts2fit400.bg_par_010off100.p};
%     end
%     cuts_list{end+1} = bg_par;
% end
bg_par = sel_info.bg_par;
cuts_list{end+1} = bg_par;
fbase_name = sel_info.fname;

```

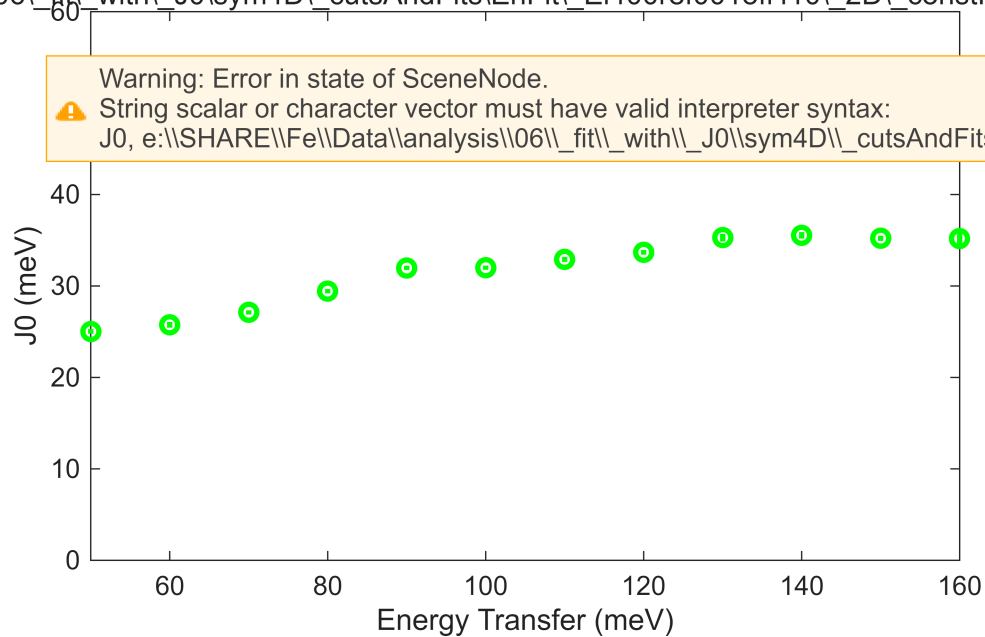
```

[fit_res_400rec,res_name] = fit_multicuts_along_direction(...
    cuts_list,fbase_name,dir_name,cut_dimensions,cut_en,dE_step,half_dE,true);

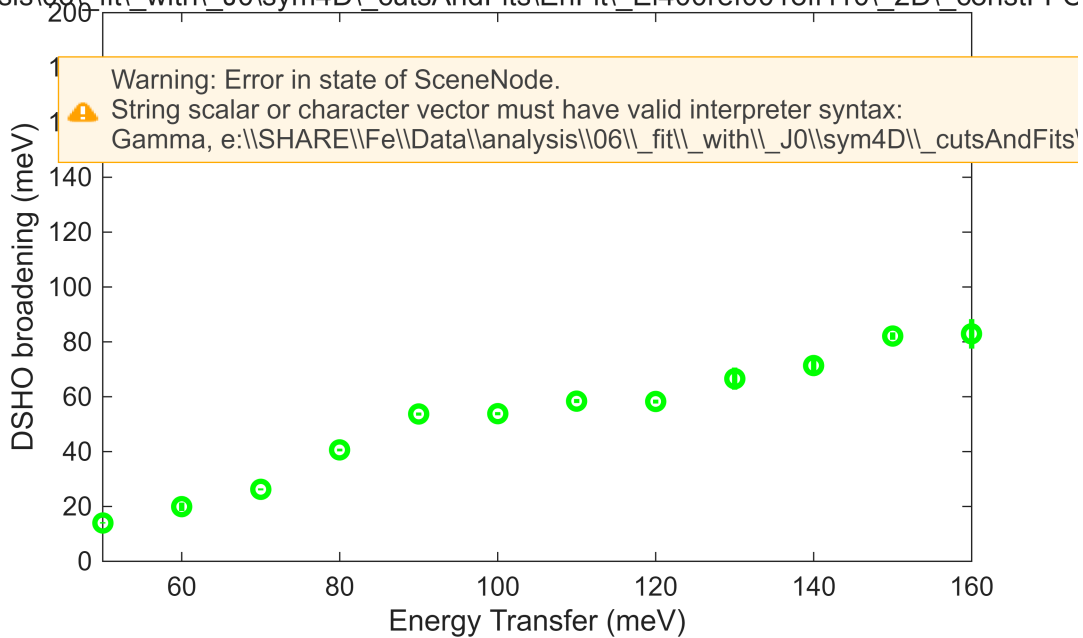
```



analysis\06\ fit_with_J0\sym4D_cutsAndFits\EnFit_Ei400ref001off110_2D_constFFCut\



analysis\06\ fit_with_J0\sym4D_cutsAndFits\EnFit_Ei400ref001off110_2D_constFFC



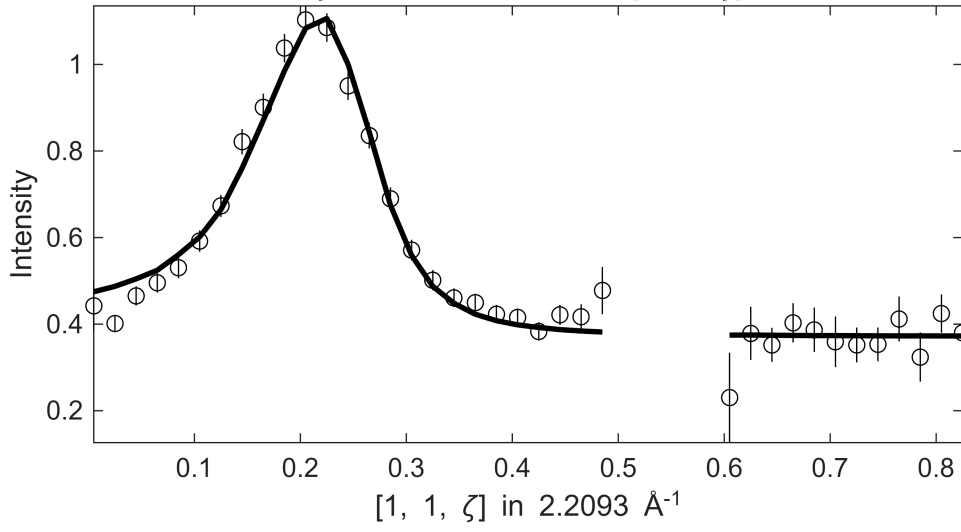
 **** En = 50±5

Fe_ei401_align_sym3.sqw

S=0.89±0.07; J0= 25±0.49; gamma=18.0982±2.38993

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 45 ≤ E ≤ 55

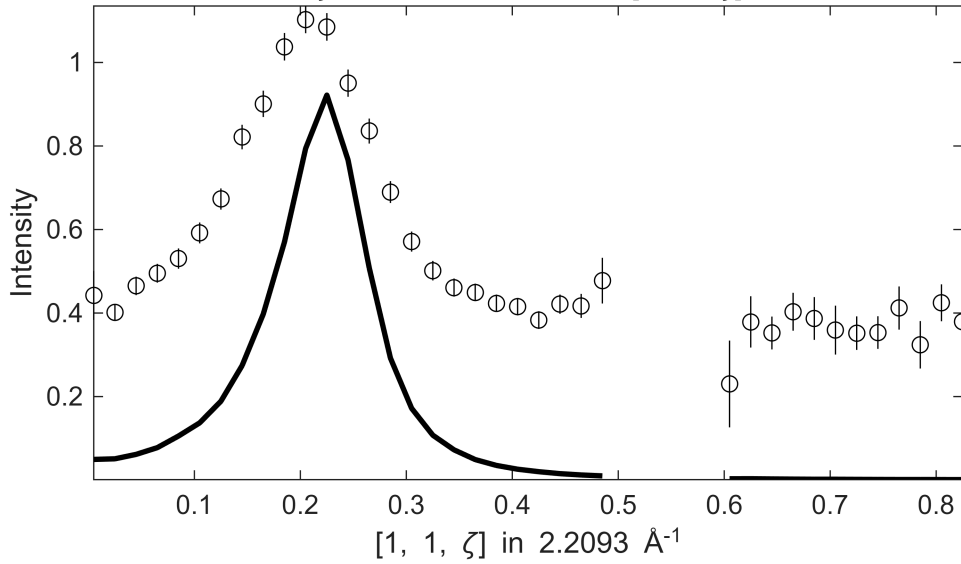
ζ=-0.005:0.02:1.015 in [1, 1, ζ]



Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 45 ≤ E ≤ 55

ζ=-0.005:0.02:1.015 in [1, 1, ζ]

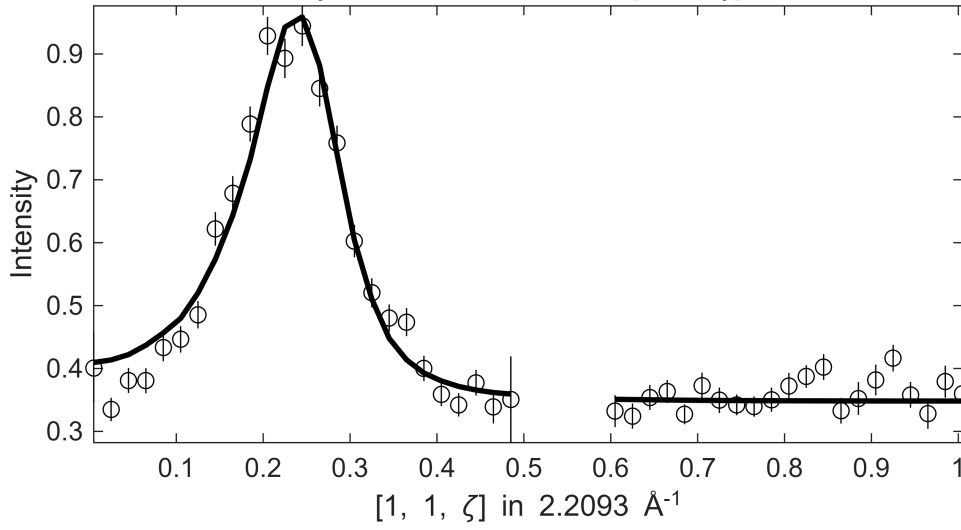


**** En = 60±5

Fe_ei401_align_sym3.sqw

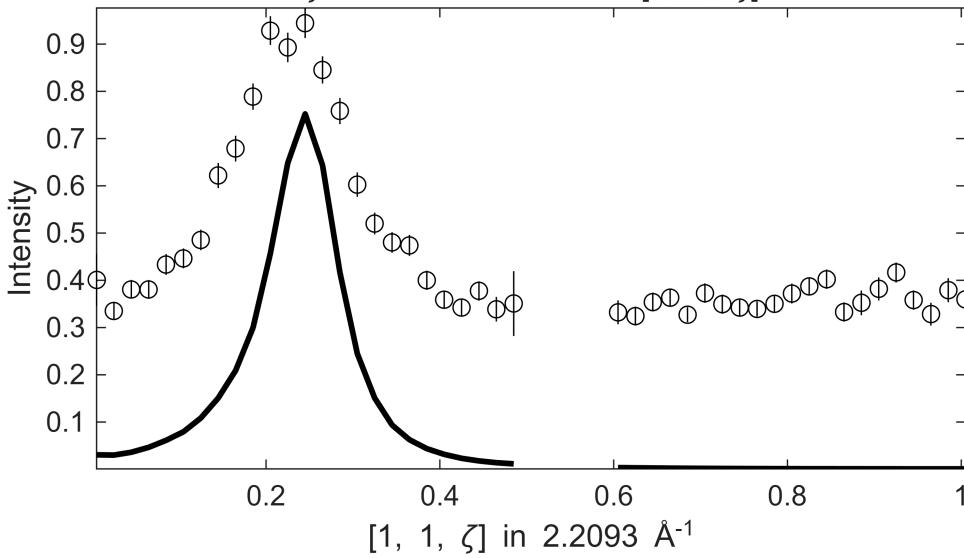
S= 0.8±0.06; J0= 26±0.56; gamma=19.6974±2.73317

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 55 ≤ E ≤ 65
ζ=-0.005:0.02:1.015 in [1, 1, ζ]



Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 55 ≤ E ≤ 65
ζ=-0.005:0.02:1.015 in [1, 1, ζ]



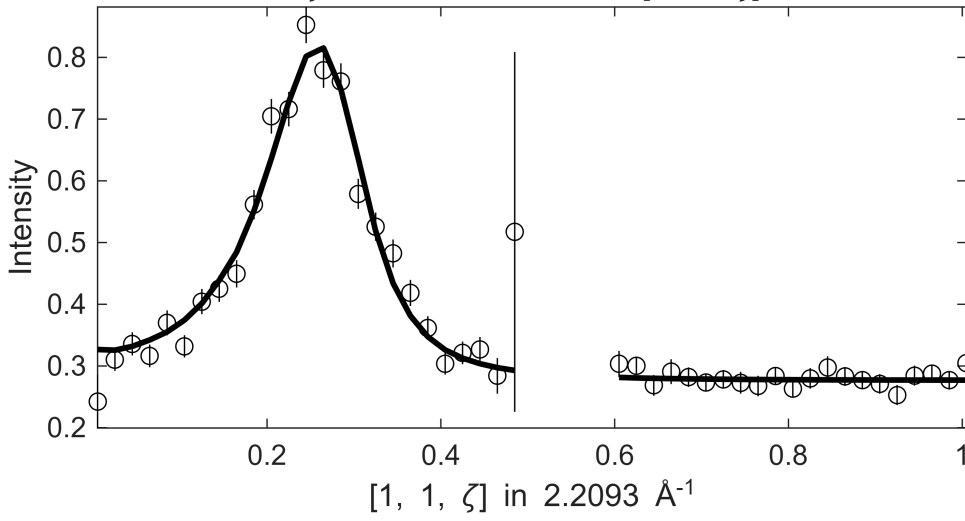
**** En = 70±5

Fe_ei401_align_sym3.sqw

S=0.84±0.018; J0= 27± 0.3; gamma=25.1576±0.473569

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 65 ≤ E ≤

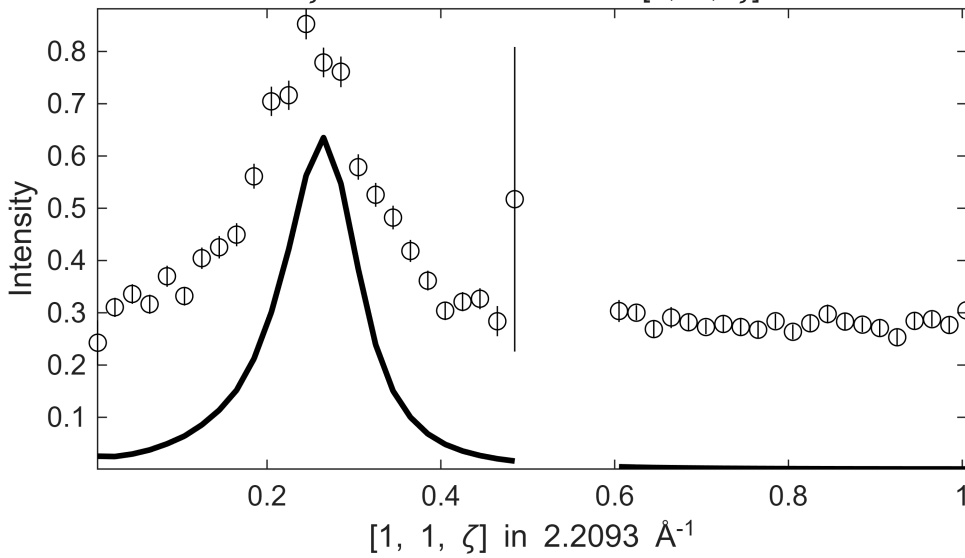
ζ=-0.005:0.02:1.015 in [1, 1, ζ]



Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 65 ≤ E ≤

ζ=-0.005:0.02:1.015 in [1, 1, ζ]



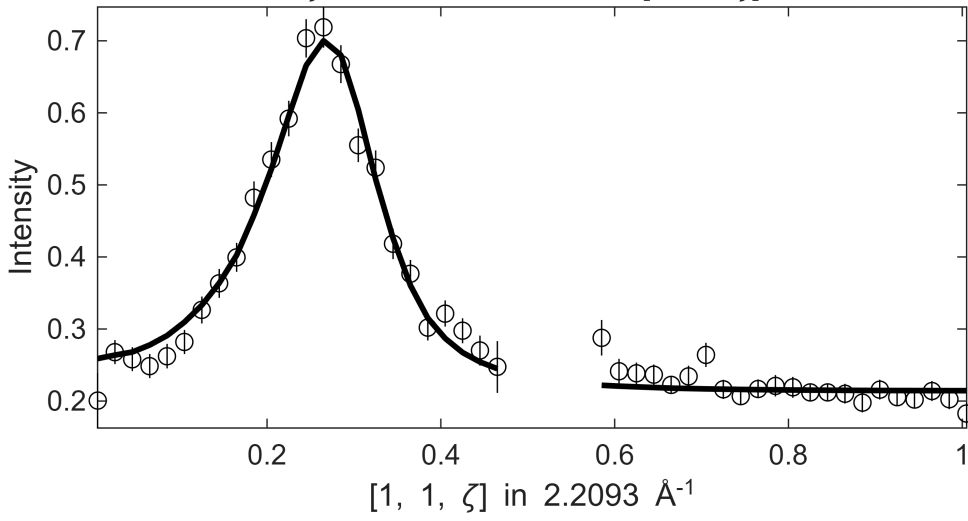
**** En = 80±5

Fe_ei401_align_sym3.sqw

S= 1 ± 0.022 ; J0= 29 ± 0.35 ; gamma= 36.2278 ± 0.666916

$-0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $75 \leq E \leq$

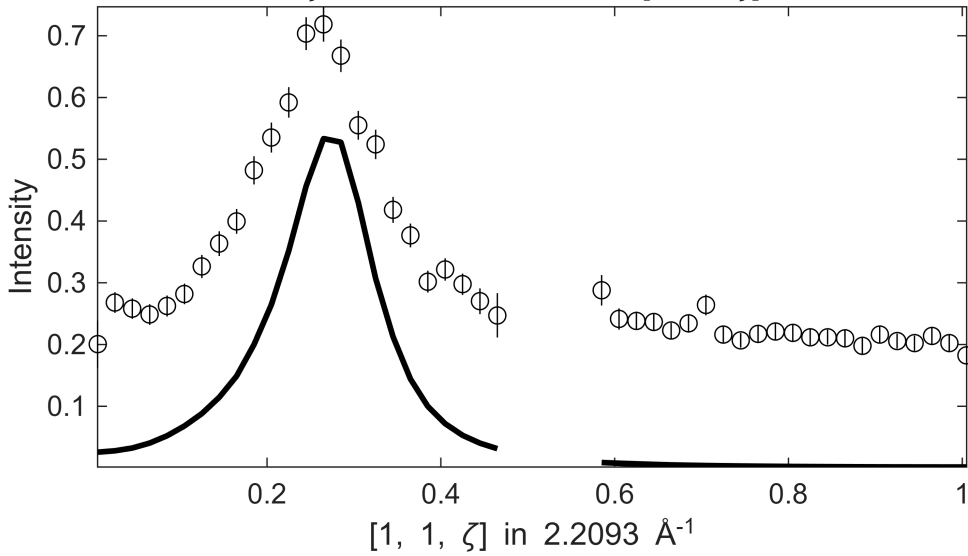
$\zeta = -0.005:0.02:1.015$ in $[1, 1, \zeta]$



Fe_ei401_align_sym3.sqw

$-0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $75 \leq E \leq$

$\zeta = -0.005:0.02:1.015$ in $[1, 1, \zeta]$



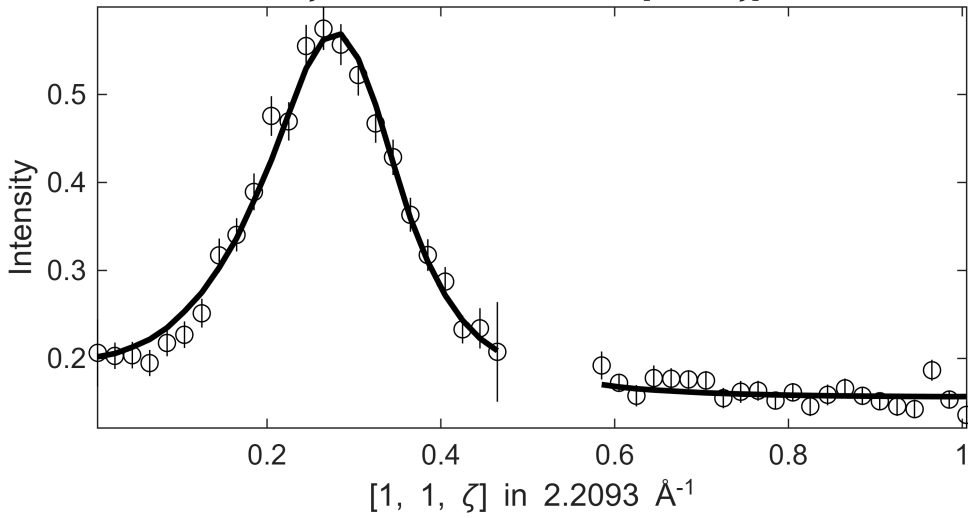
**** En = 90 ± 5

Fe_ei401_align_sym3.sqw

S= 1.3±0.025; J0= 33±0.36; gamma=62.9596±0.946298

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 85 ≤ E ≤

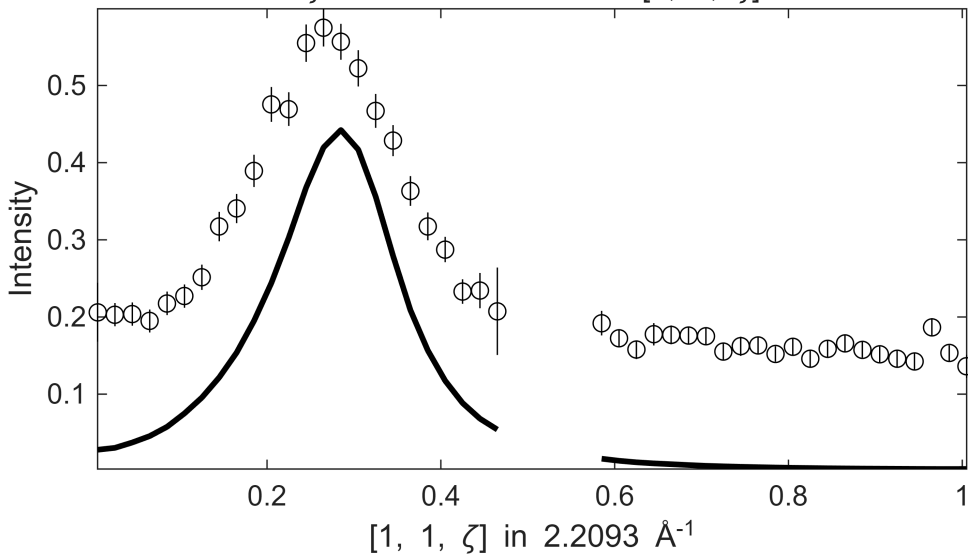
ζ=-0.005:0.02:1.015 in [1, 1, ζ]



Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 85 ≤ E ≤

ζ=-0.005:0.02:1.015 in [1, 1, ζ]



**** En = 100±5

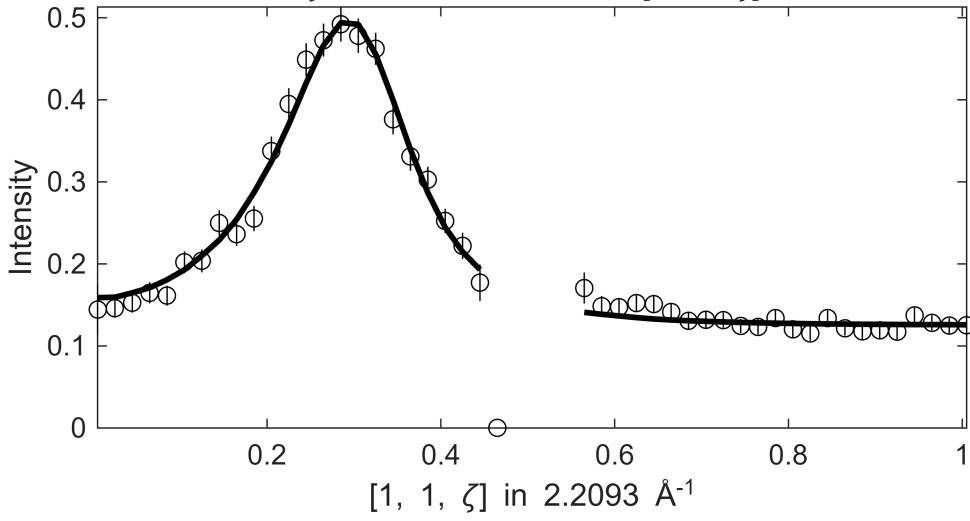
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.9608

Fe_ei401_align_sym3.sqw

S= 1.1±0.02; J0= 33±0.33; gamma=57.5853±0.901629

0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 95 ≤ E ≤

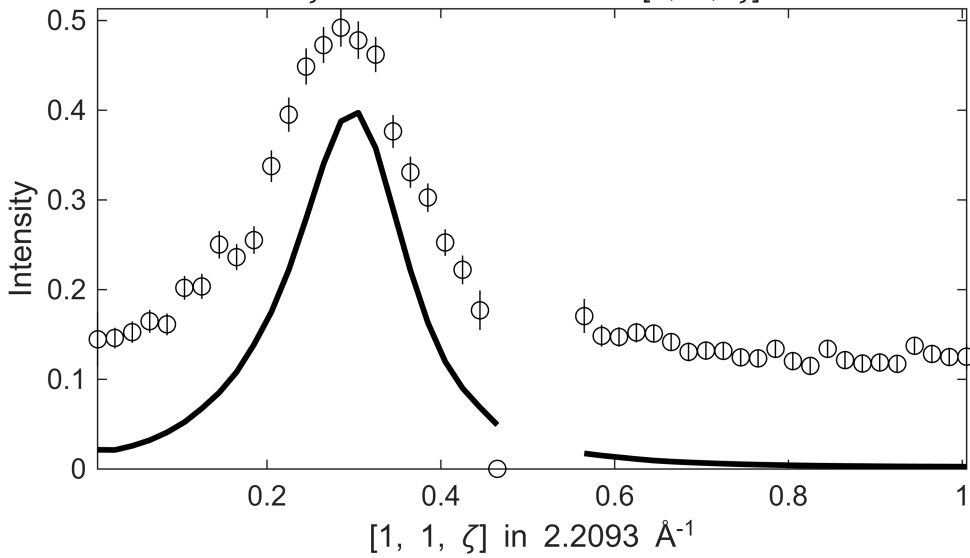
ζ=-0.005:0.02:1.015 in [1, 1, ζ]



Fe_ei401_align_sym3.sqw

0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 95 ≤ E ≤

ζ=-0.005:0.02:1.015 in [1, 1, ζ]



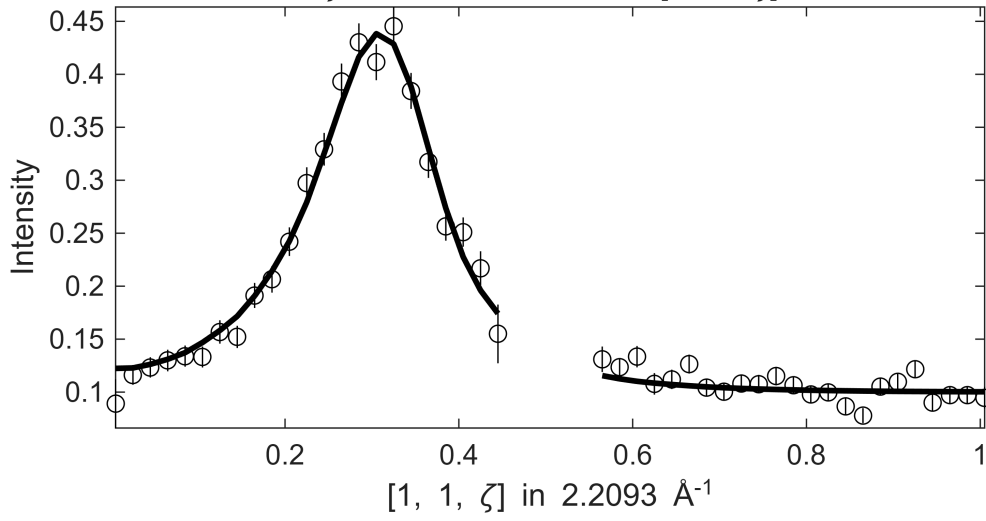
**** En = 110±5

Fe_ei401_align_sym3.sqw

S=0.97±0.021; J0= 32±0.37; gamma=52.592±0.957739

0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 105 ≤ E ≤

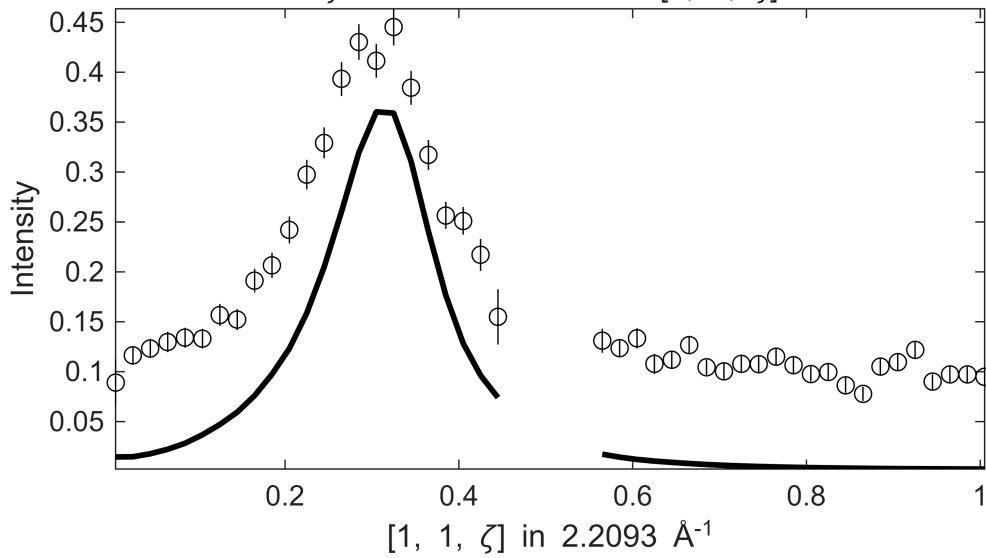
ζ=-0.005:0.02:1.015 in [1, 1, ζ]



Fe_ei401_align_sym3.sqw

0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 105 ≤ E ≤

ζ=-0.005:0.02:1.015 in [1, 1, ζ]



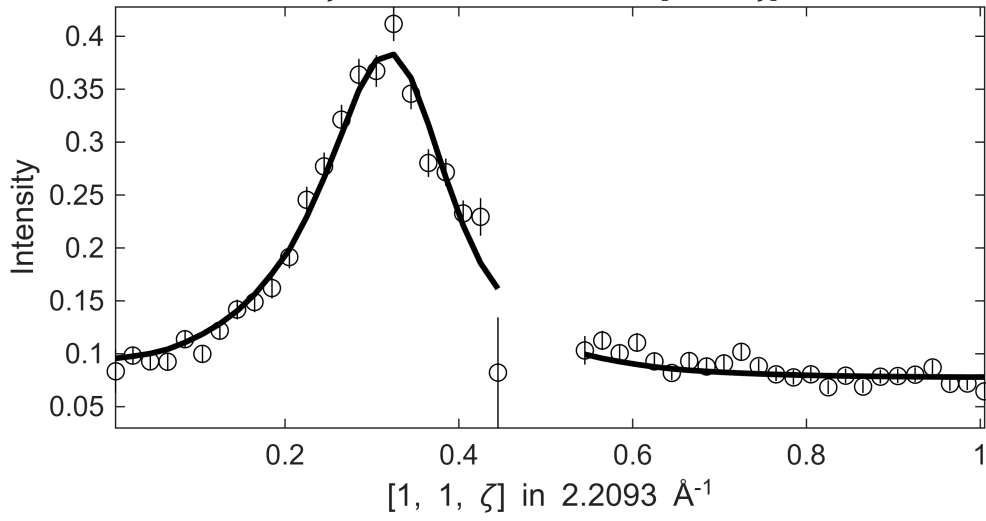
**** En = 120±5

Fe_ei401_align_sym3.sqw

S= 1 ± 0.064 ; J0= 34 ± 0.65 ; gamma= 61.1136 ± 5.85448

$0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $115 \leq E \leq$

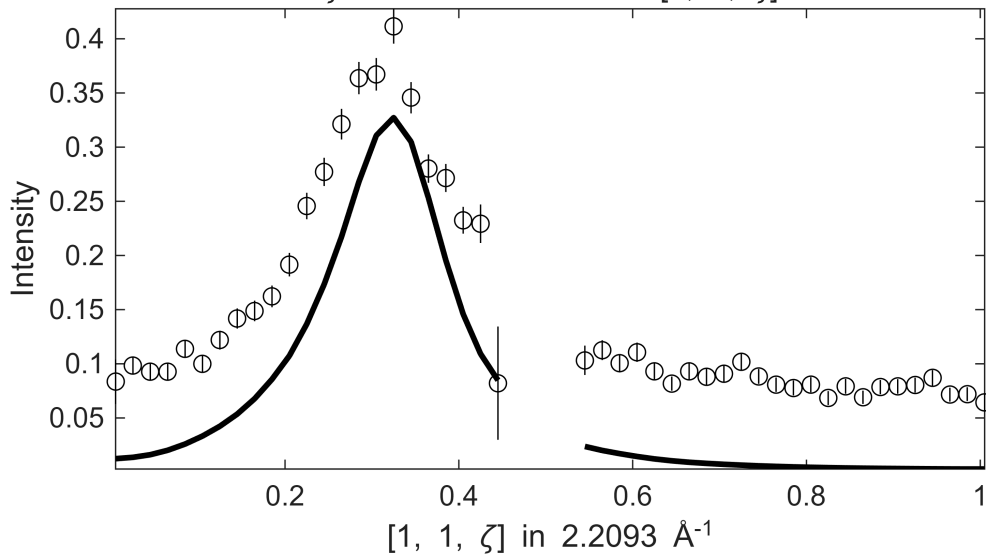
$\zeta = -0.005:0.02:1.015$ in $[1, 1, \zeta]$



Fe_ei401_align_sym3.sqw

$0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $115 \leq E \leq$

$\zeta = -0.005:0.02:1.015$ in $[1, 1, \zeta]$



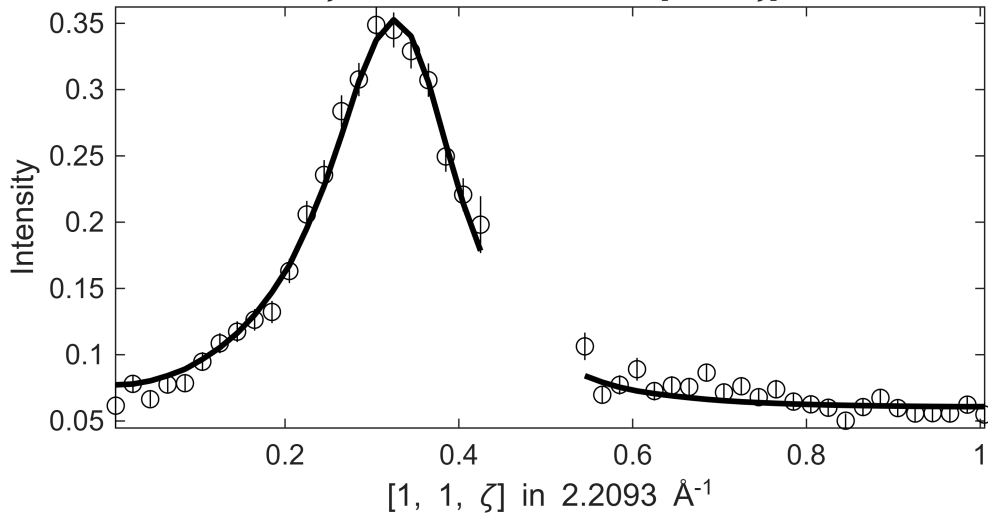
**** En = 130 ± 5

Fe_ei401_align_sym3.sqw

S= 1±0.021; J0= 35± 0.4; gamma=65.3038±1.18136

0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 125 ≤ E ≤

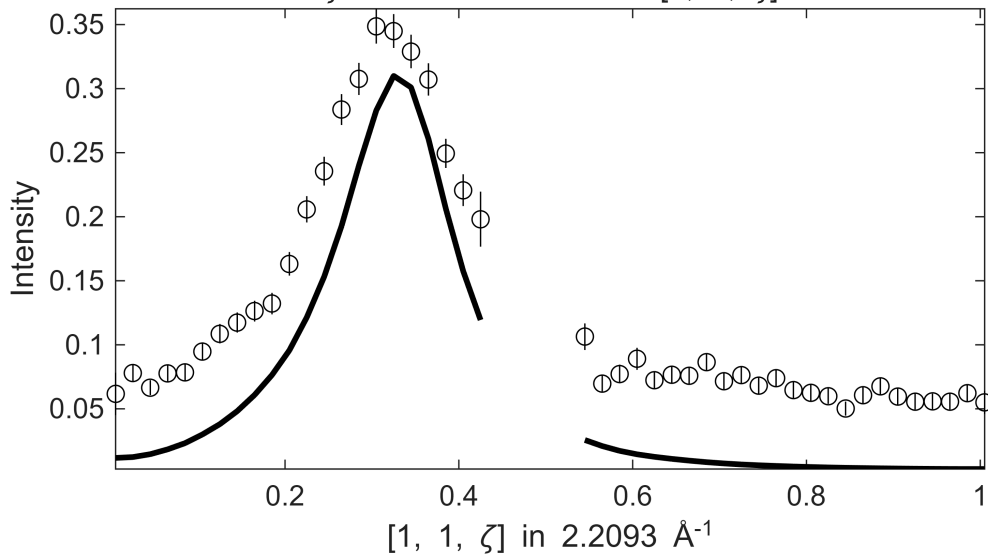
ζ=-0.005:0.02:1.015 in [1, 1, ζ]



Fe_ei401_align_sym3.sqw

0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 125 ≤ E ≤

ζ=-0.005:0.02:1.015 in [1, 1, ζ]

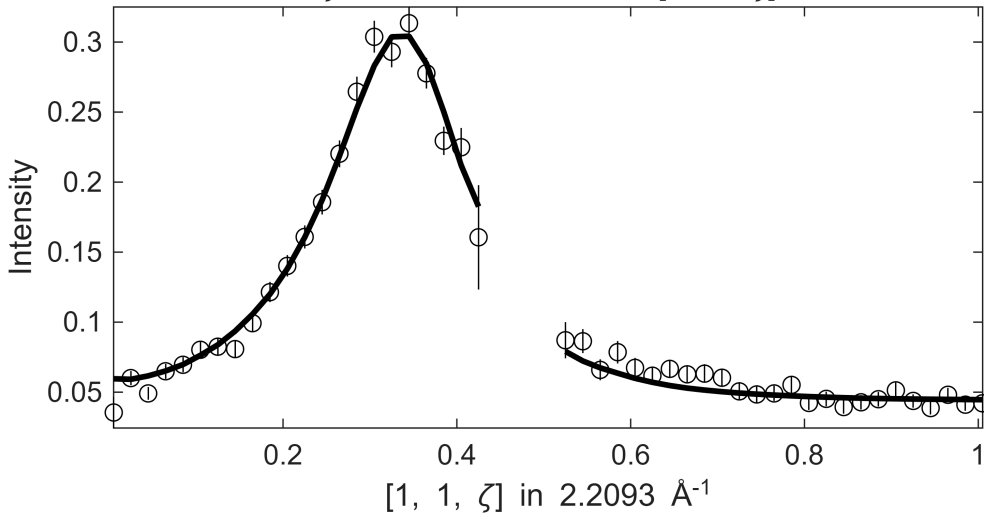


**** En = 140±5

Fe_ei401_align_sym3.sqw

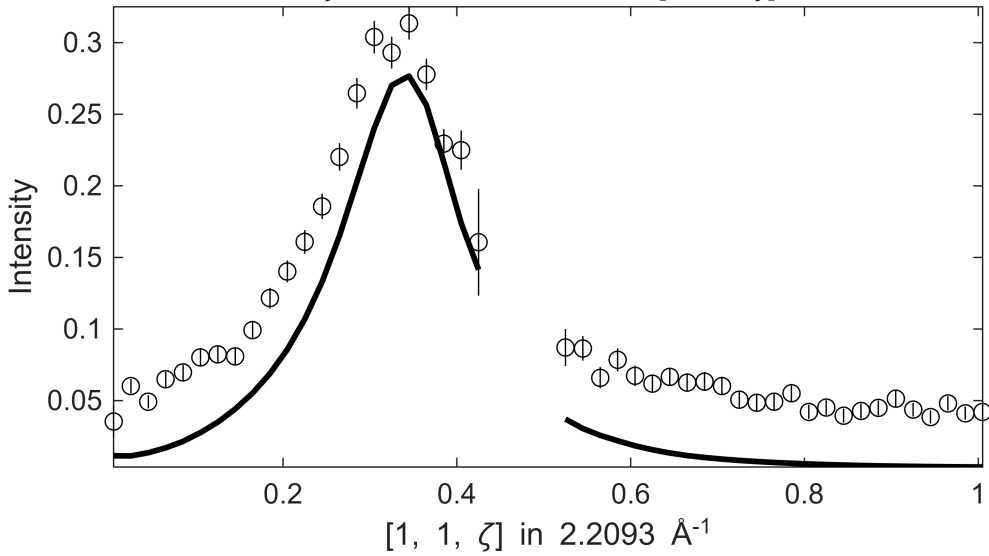
S= 1.1±0.026; J0= 37±0.43; gamma=76.8545±1.50782

$0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $135 \leq E \leq$
 $\zeta=-0.005:0.02:1.015$ in $[1, 1, \zeta]$



Fe_ei401_align_sym3.sqw

$0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $135 \leq E \leq$
 $\zeta=-0.005:0.02:1.015$ in $[1, 1, \zeta]$



**** En = 150±5

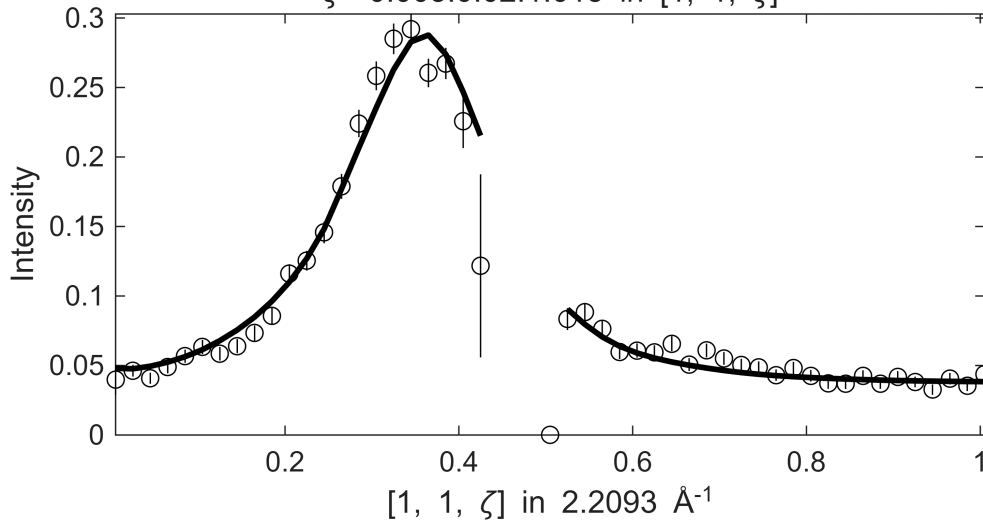
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.9608

Fe_ei401_align_sym3.sqw

S= 1.1±0.065; J0= 35±0.69; gamma=82.1373±6.74209

$0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $145 \leq E \leq$

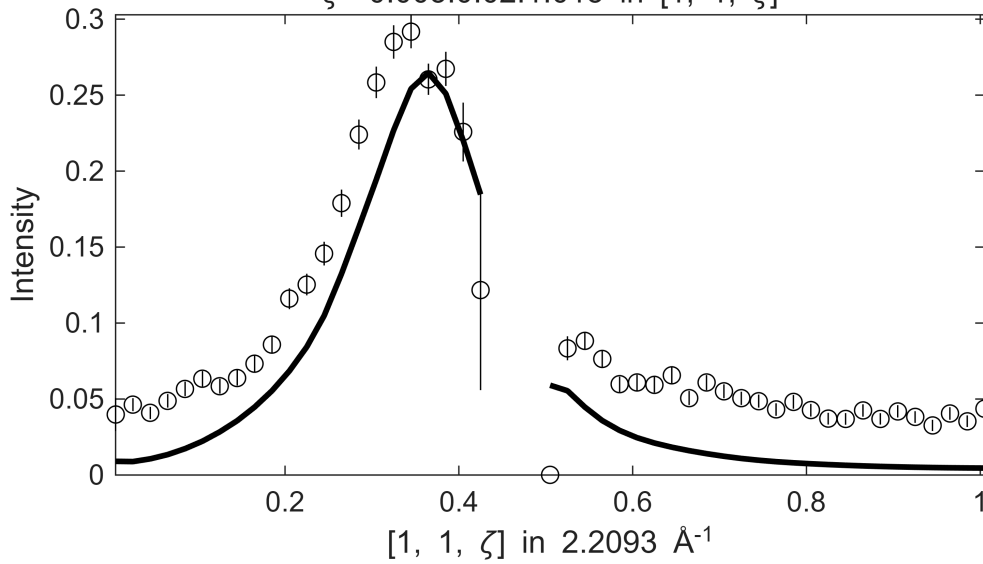
$\zeta=-0.005:0.02:1.015$ in $[1, 1, \zeta]$



Fe_ei401_align_sym3.sqw

$0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $145 \leq E \leq$

$\zeta=-0.005:0.02:1.015$ in $[1, 1, \zeta]$

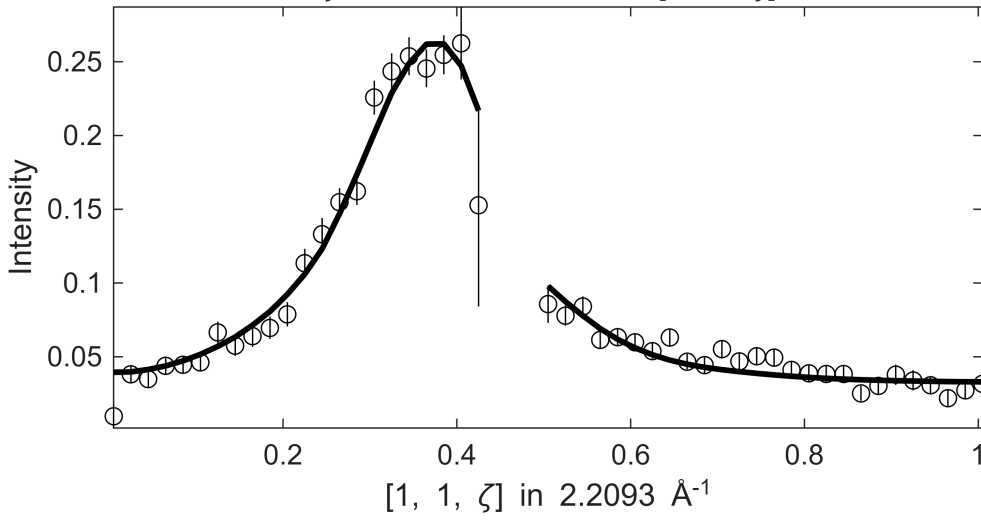


**** En = 160±5

Fe_ei401_align_sym3.sqw

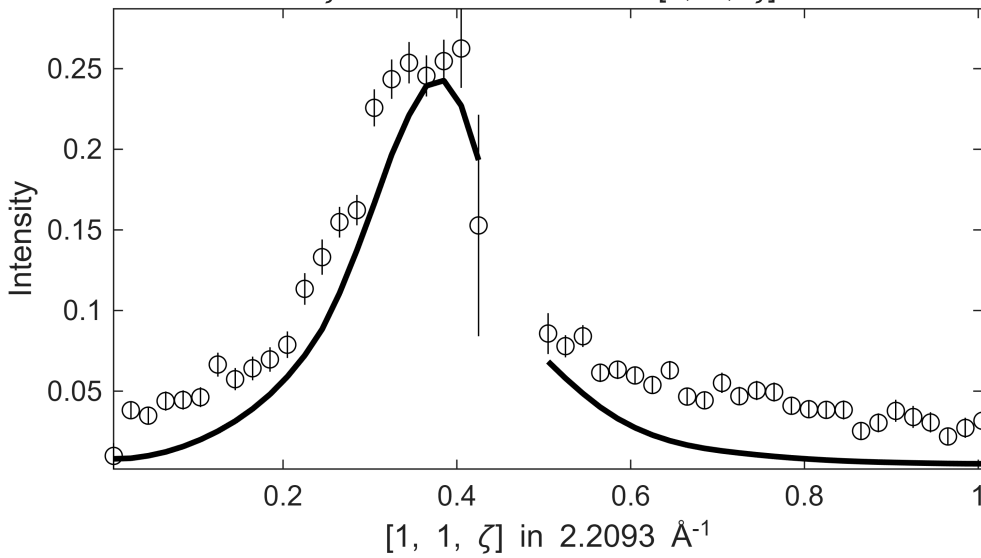
S= 1.1±0.049; J0= 35±0.48; gamma=85.7972±3.26124

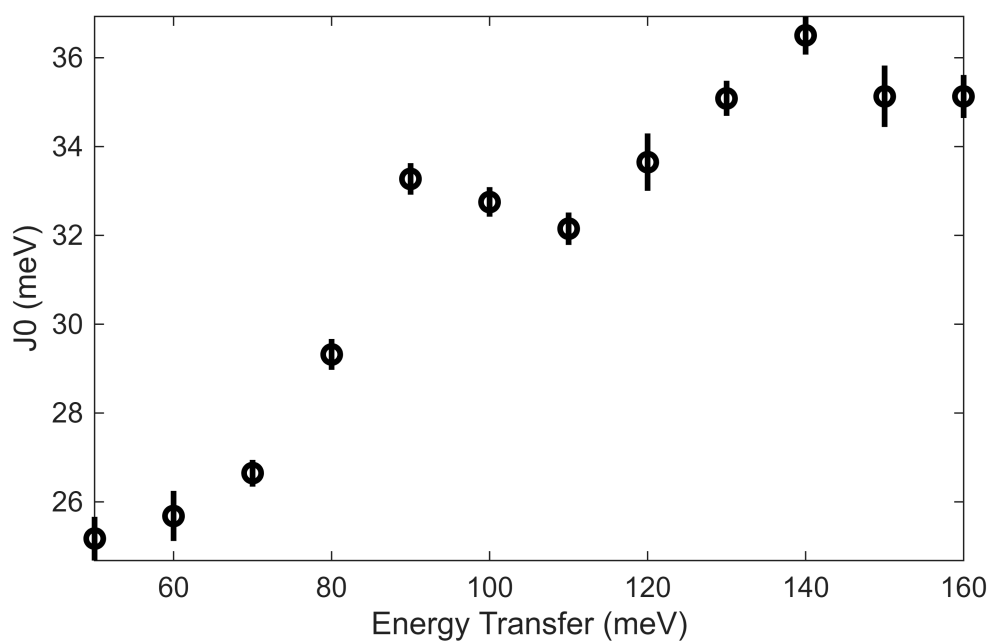
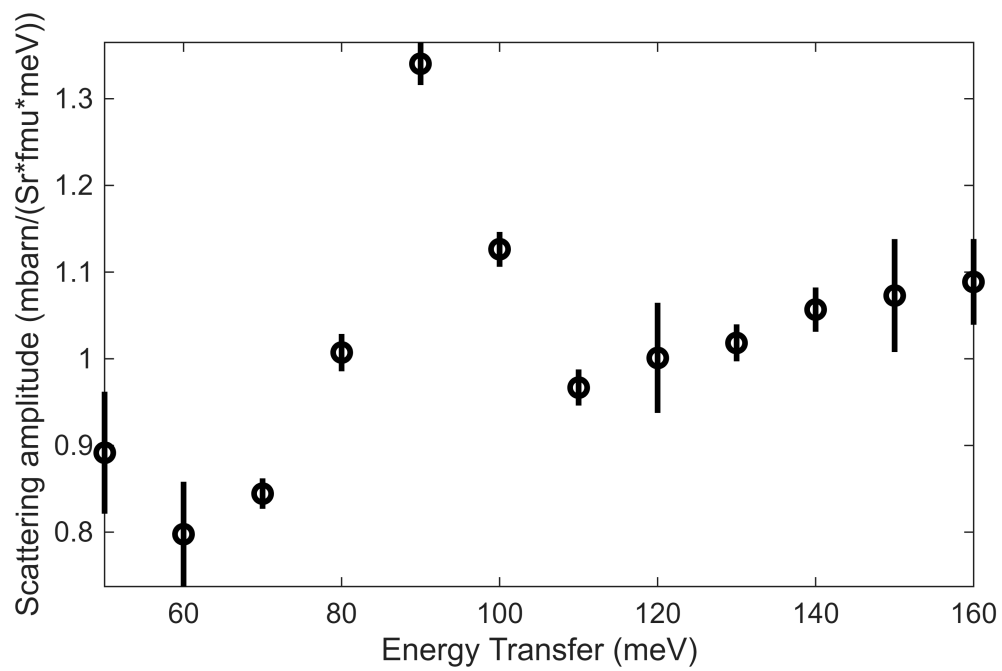
$0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $155 \leq E \leq$
 $\zeta=-0.005:0.02:1.015$ in $[1, 1, \zeta]$

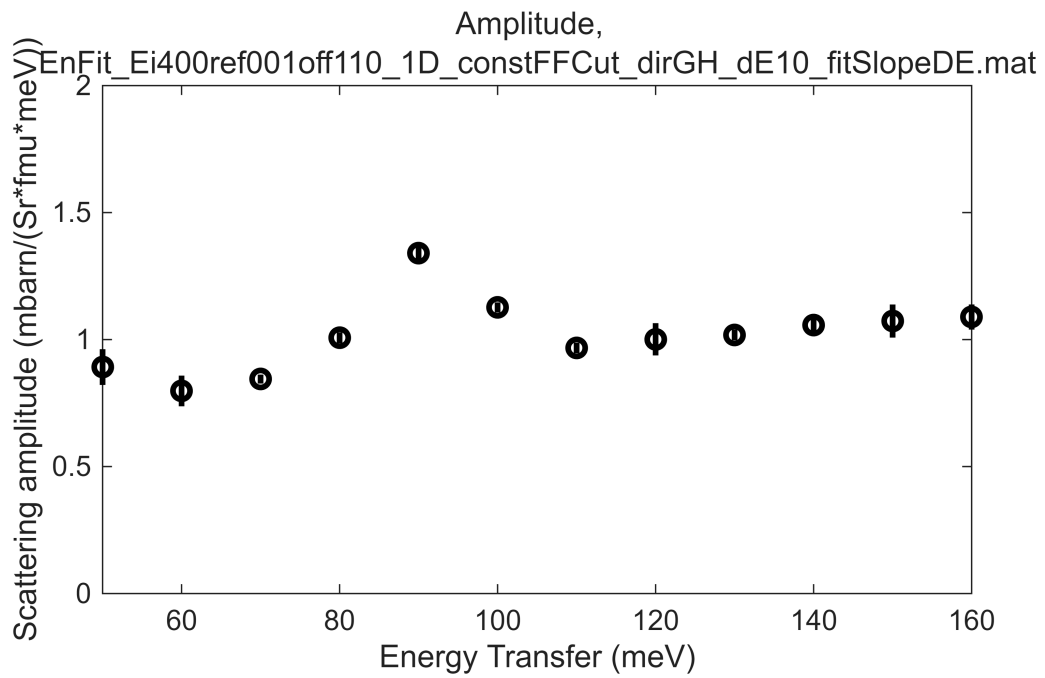
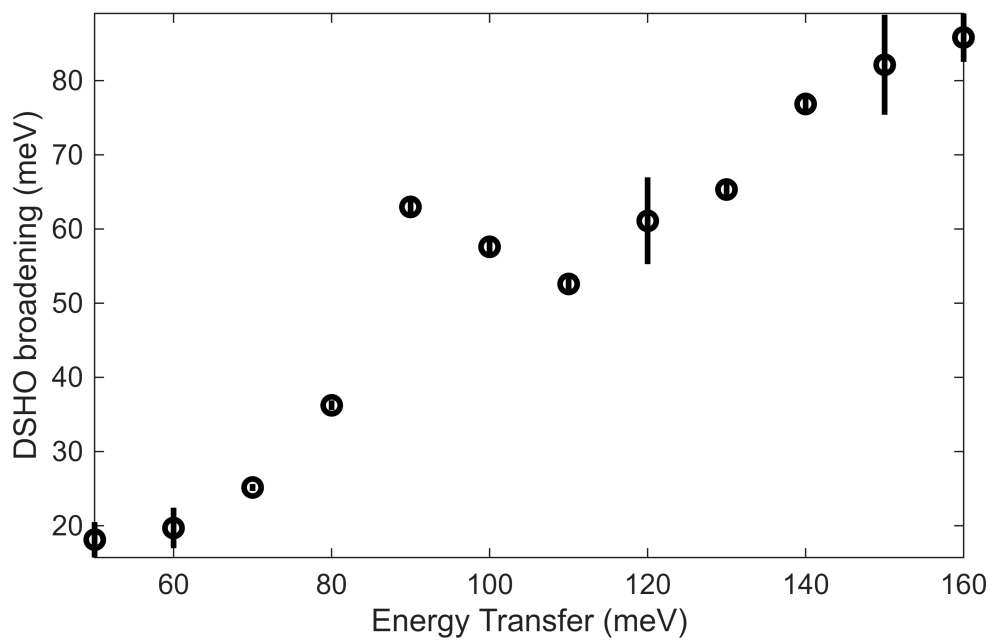


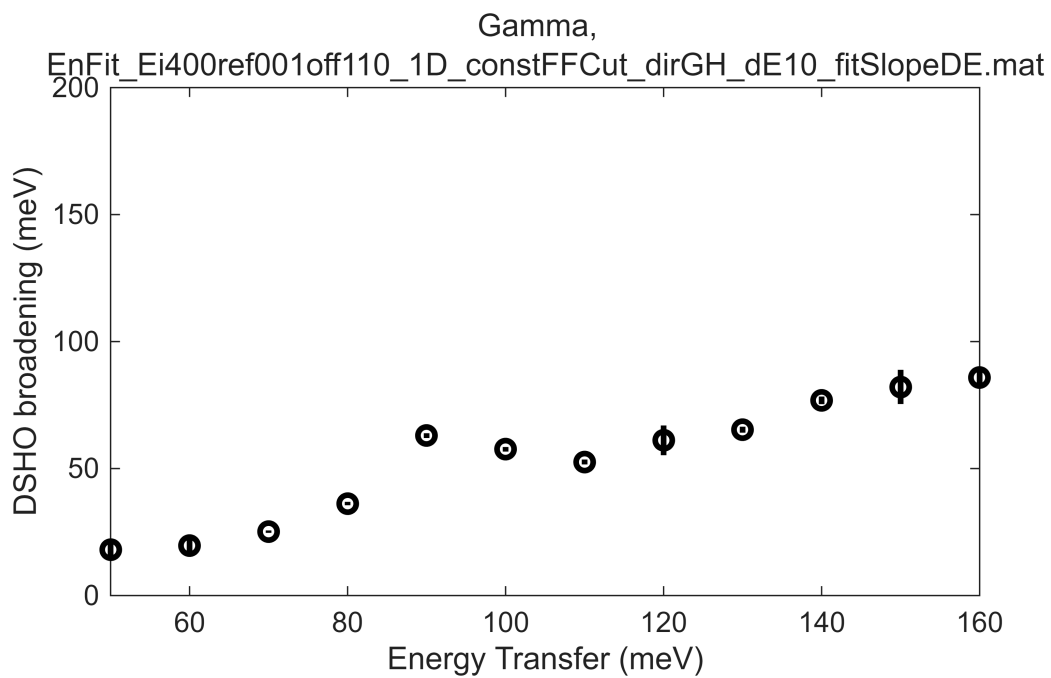
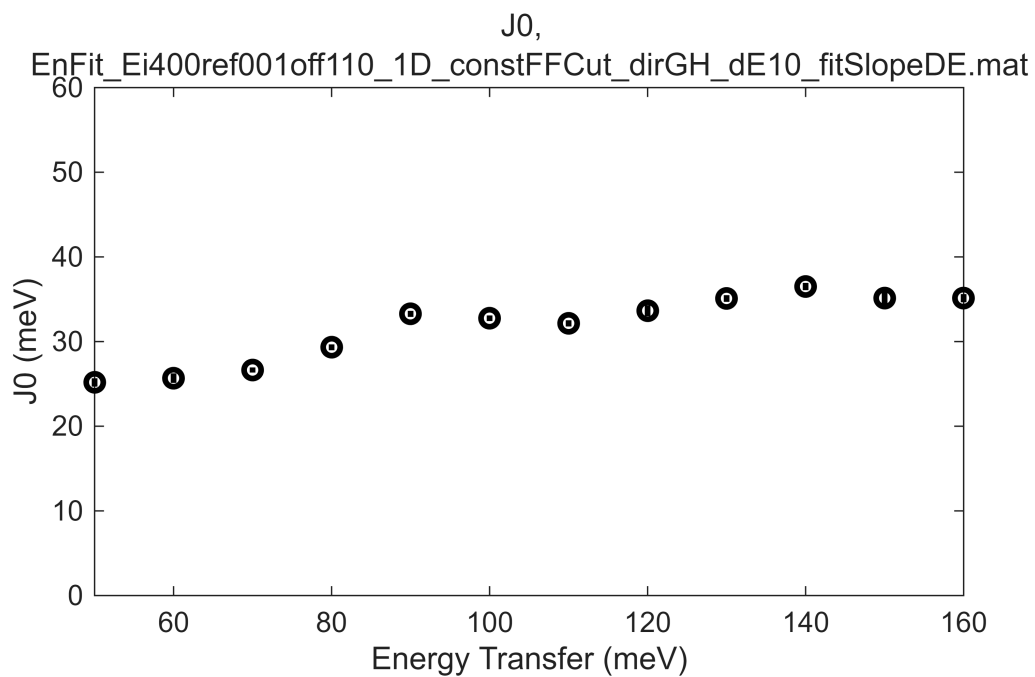
Fe_ei401_align_sym3.sqw

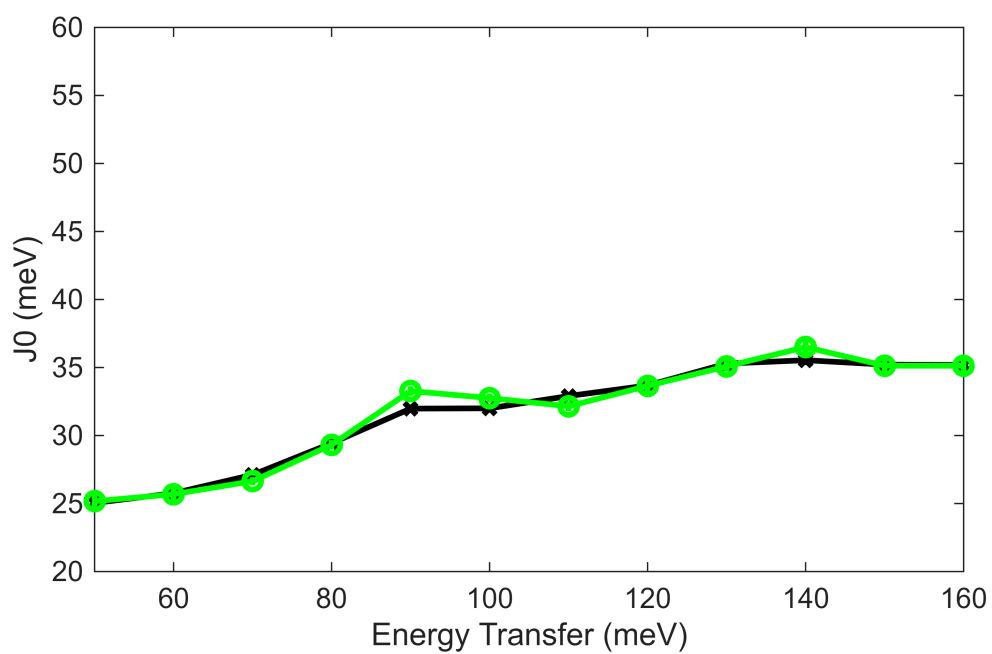
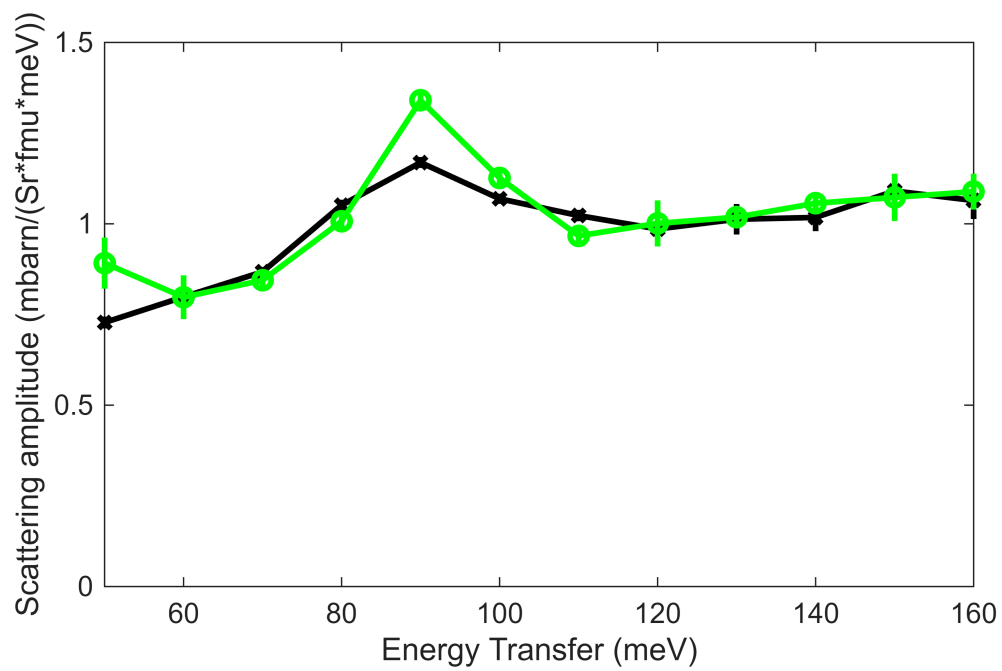
$0.1 \leq \xi \leq 0.1$ in $[1+\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1+\eta, 0]$, $155 \leq E \leq$
 $\zeta=-0.005:0.02:1.015$ in $[1, 1, \zeta]$

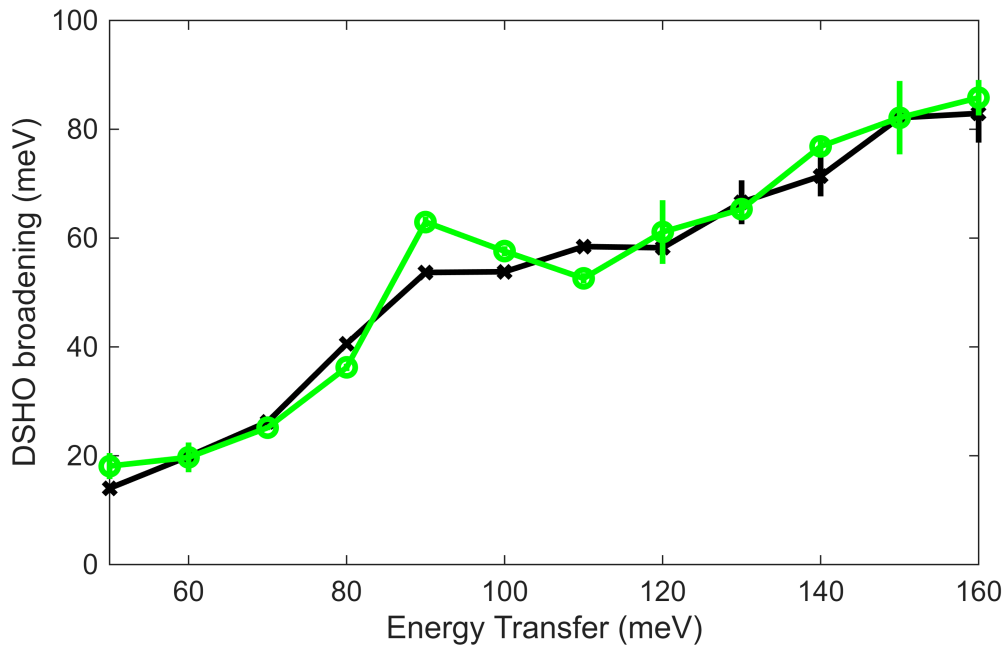












Warning: Error in state of SceneNode.

String scalar or character vector must have valid interpreter syntax:

Amplitude,

e:\SHARE\FE\Data\analysis\06_fit\with\J0\sym4D\cutsAndFits\EnFit\Ei400ref001off110_2D_constFFCut_d
irGH_dE20_fitSlopeDE.mat

Warning: Error in state of SceneNode.

String scalar or character vector must have valid interpreter syntax:

J0,
e:\SHARE\FE\Data\analysis\06_fit\with\J0\sym4D\cutsAndFits\EnFit\Ei400ref001off110_2D_constFFCut_d
irGH_dE20_fitSlopeDE.mat

Warning: Error in state of SceneNode.

String scalar or character vector must have valid interpreter syntax:

Gamma,

e:\SHARE\FE\Data\analysis\06_fit\with\J0\sym4D\cutsAndFits\EnFit\Ei400ref001off110_2D_constFFCut_d
irGH_dE20_fitSlopeDE.mat

```
%cl,'En_cuts800',dir_name,cut_en,dE_step,half_dE);
```

```
beep
```

```
beep
```

```
beep
```

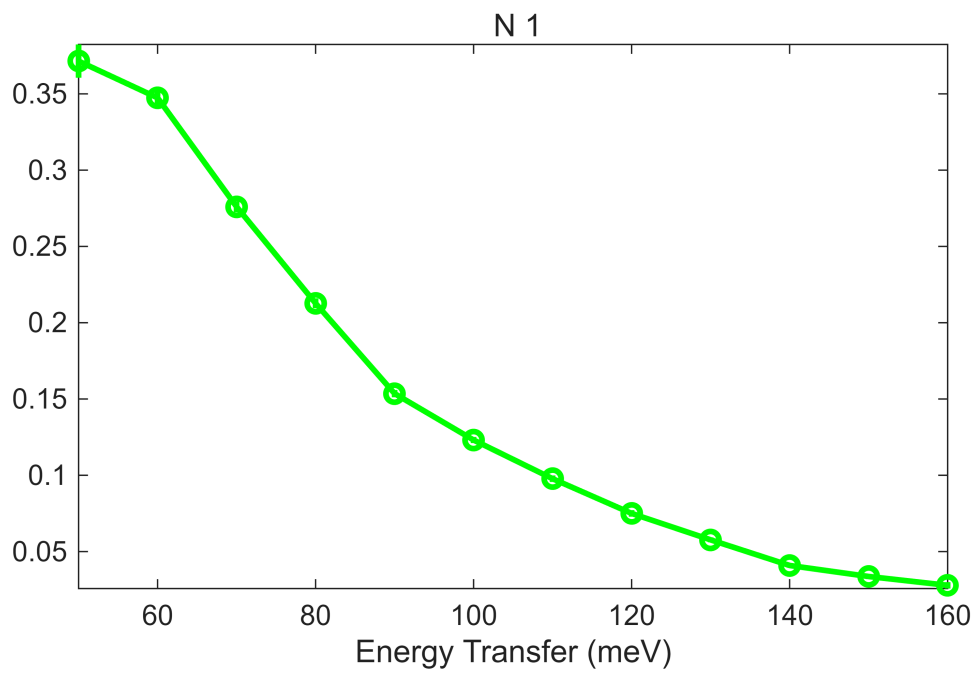
```
fit_res_400 = fit_res_400rec;
```

```
bg_data = extract_bg_par(fit_res_400.all_fit_par,[1,2]);
```

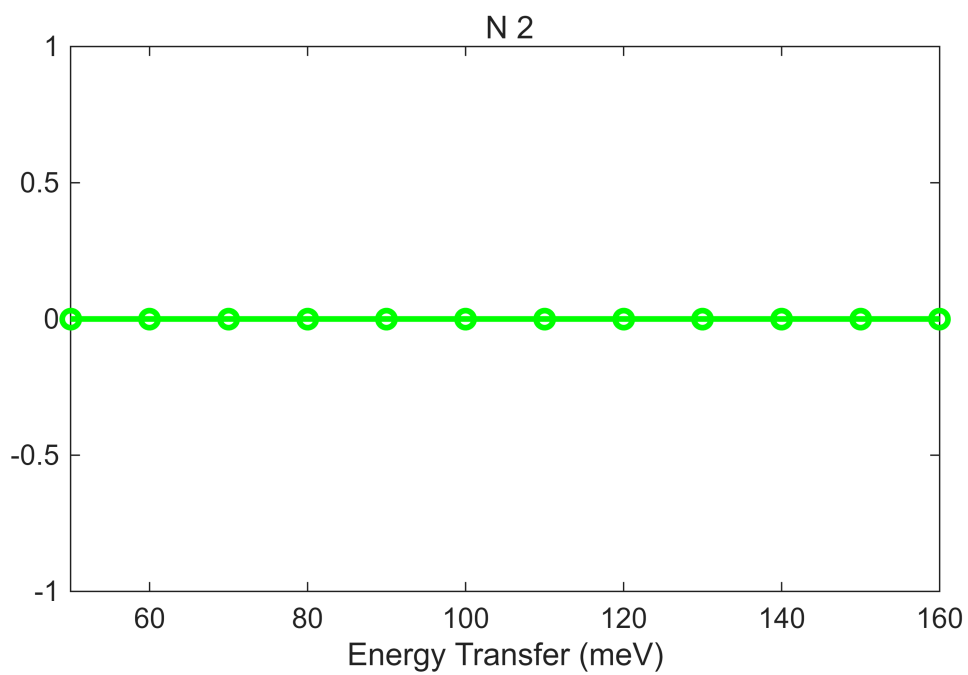
```
% original slope parameters
```

```
pd(bg_data(1));lx(fit_res_400.all_fit_par(1).en,fit_res_400.all_fit_par(end).en);
```

```
keep_figure;
```



```
pd(bg_data(2));lx(fit_res_400.all_fit_par(1).en,fit_res_400.all_fit_par(end).en);
keep_figure
```



```
fit_resEi400 = fit_res_400rec;
save(res_name,"fit_resEi400")
```